

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 27

Explore and Remove a Default VPC

Amazon VPC | Subnets | Route Tables | Internet Gateway | Cleanup

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ENVIRONMENT	Personal AWS Lab
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STATUS	Complete

Executive Summary

Created a default **Amazon VPC** in the workload account for networking exploration, reviewed its core components, and deleted it after testing. The default VPC contained four subnets across **us-west-2a**, **us-west-2b**, **us-west-2c**, and **us-west-2d**, one route table, and one Internet Gateway. The VPC was removed after the lab so the environment would not retain an unnecessary default public network.

SCOPE

This was an exploration lab, not a production network build. The default VPC was intentionally deleted after review because the next lab will build a replacement VPC from scratch.

Business Problem

Before designing custom cloud networks, it is important to understand the default VPC that AWS can create for each region. Default VPCs are convenient, but they can become a security risk if resources are deployed there unintentionally. This lab established baseline VPC understanding while also practicing cleanup of unneeded network infrastructure.

Objectives

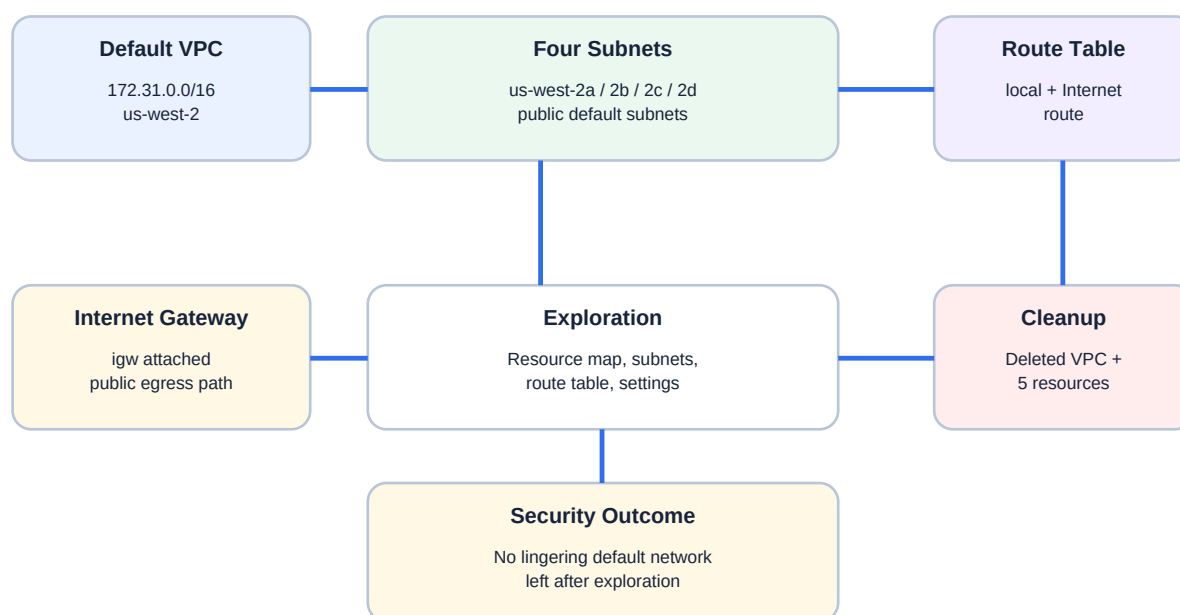
- Create or restore the default VPC for temporary exploration.
- Review the VPC details, including CIDR range, DNS settings, route table, network ACL, and Internet Gateway.
- Inspect the VPC resource map and identify the relationship between VPC, subnets, route table, and Internet Gateway.
- Review the four default subnets across the us-west-2 Availability Zones.
- Delete the default VPC and dependent resources after the lab to prevent accidental future use.
- Document the exercise as an AWS networking foundation project.

Environment

Cloud Provider	Amazon Web Services (AWS)
Primary Service	Amazon VPC
Account Context	Workload/test account
Region	us-west-2 / Oregon
VPC Type	Default VPC created for lab exploration
IPv4 CIDR	172.31.0.0/16
Availability Zones	us-west-2a, us-west-2b, us-west-2c, us-west-2d

Subnets	4 default subnets
Route Tables	1 main route table
Network Connections	1 Internet Gateway
Cleanup	VPC deleted along with 5 dependent resources
Evidence Redaction	Account owner ID redacted from VPC details screenshot.

Architecture



Explored the default public VPC primitives, then deleted the lab VPC and dependent resources.

Figure 1 - Default VPC exploration and cleanup model.

Implementation Summary

1. Created a default VPC in us-west-2 for testing and exploration.
2. Opened the VPC details page and confirmed the VPC was available.
3. Reviewed DNS resolution, DNS hostnames, main network ACL, DHCP option set, main route table, IPv4 CIDR, and Internet Gateway attachment.
4. Used the resource map to trace the relationship from VPC to subnets, route table, and network connection.
5. Confirmed the default VPC contained four subnets distributed across us-west-2a, us-west-2b, us-west-2c, and us-west-2d.

6. Reviewed that the route table and Internet Gateway made the default subnets public by default.

7. Deleted the VPC after the lab and confirmed deletion of the VPC and five dependent resources.

VPC Component Map

VPC	Default VPC with 172.31.0.0/16	Acts as the virtual network boundary for AWS resources.
Subnets	4 subnets across 4 AZs	Divide the VPC CIDR across Availability Zones for resource placement.
Route table	1 main route table	Controls how subnet traffic routes locally or toward external networks.
Internet Gateway	1 IGW attached	Provides a route target for Internet-bound traffic.
DNS settings	DNS resolution and hostnames enabled	Supports internal name resolution and DNS hostname behavior.
Cleanup	VPC and dependent resources deleted	Prevents accidental resource placement in the default public network.

Evidence

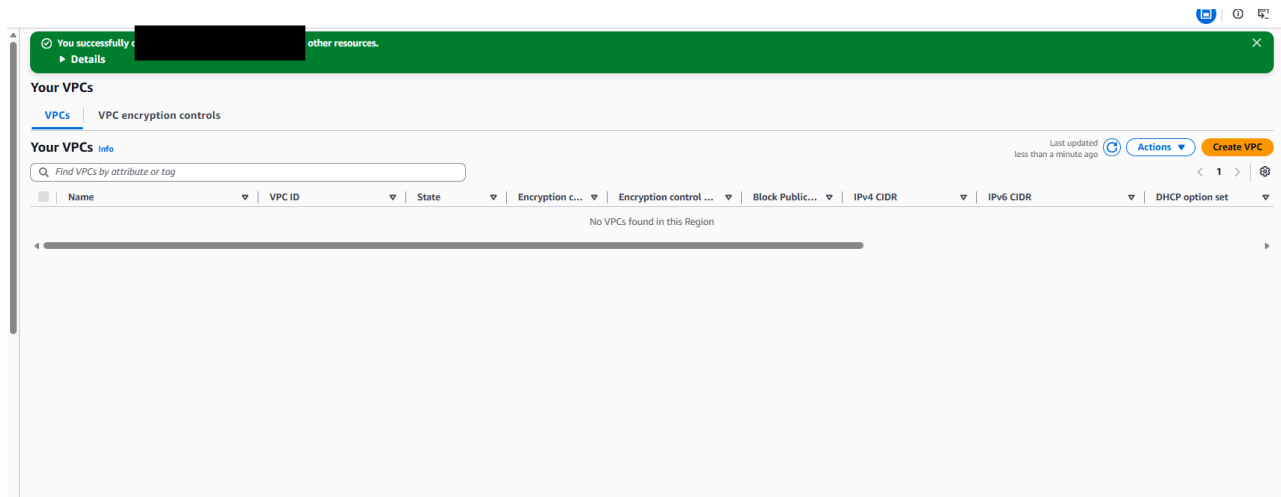
Figure 2 - Default VPC Created and Explored

The screenshot displays the AWS Management Console interface for a VPC. At the top, a green notification bar states "You successfully created vpc-0ba290...". Below this, the "Details" tab is active, showing various configuration options. The "State" is "Available". The "DNS resolution" is "Enabled". The "Main network ACL" is "Default". The "IPv6 CIDR (Network border group)" is "None". The "Encryption control ID" is "None". The "Block Public Access" is "Off". The "DHCP option set" is "Default". The "DNS hostnames" are "Enabled". The "Main route table" is "Default". The "IPv6 pool" is "None". The "Owner ID" is "REDACTED".

Below the details, the "Resource map" tab is active, showing a visual representation of the VPC resources. The map includes a "VPC" box, a "Subnets (4)" box, a "Route tables (1)" box, and a "Network Connections (1)" box. The "Subnets (4)" box lists four subnets: "us-west-2a", "us-west-2b", "us-west-2c", and "us-west-2d". The "Route tables (1)" box shows a single route table. The "Network Connections (1)" box shows a single connection.

The VPC evidence shows successful creation of a default VPC in us-west-2. The resource map shows four subnets, one route table, and one Internet Gateway. DNS resolution and DNS hostnames are enabled, and the IPv4 CIDR is 172.31.0.0/16. The AWS account owner ID was redacted.

Figure 3 - Default VPC Deleted After Testing



The cleanup evidence shows the default VPC was successfully deleted along with five other resources. The VPC list then shows no VPCs found in the region, confirming the temporary test VPC was removed.

Security Analysis

Default VPCs are convenient because resources can be launched without designing a network first. That same convenience creates risk when users accidentally deploy resources into a public-by-default network.

The route table and Internet Gateway relationship is the key reason this default VPC should be treated carefully. A 0.0.0.0/0 route to an Internet Gateway creates public-subnet behavior.

Deleting the default VPC after exploration reduces the chance of accidental workload deployment into an unmanaged default network.

The lab did not create a production network. A production VPC should have deliberate CIDR planning, public and private subnet separation, route-table segmentation, logging, tagging, and controlled ingress and egress.

Lessons Learned

A VPC is a software-defined virtual network, not a traditional physical network segment.

The default VPC is composed of several primitives: VPC CIDR, subnets, route tables, network ACLs, and an Internet Gateway.

Subnets are placed into Availability Zones, which supports future resilience planning.

Route tables define whether subnet traffic stays local, routes to the Internet, or routes through another target.

Cleanup is part of secure lab work. Temporary resources should be deleted when their purpose is complete.

Production Improvements

- Build a custom VPC instead of relying on default VPCs.
- Use separate public and private subnets across at least two Availability Zones.

- Apply a consistent IP addressing plan before deploying workloads.
- Enable VPC Flow Logs for visibility into network activity.
- Tag VPCs, subnets, route tables, and Internet gateways consistently.
- Use security groups and network architecture rather than default-open assumptions.
- Document route-table purpose and subnet classification for every production VPC.

Skills Demonstrated

Amazon VPC	Default VPC creation	Resource cleanup
Subnets and AZs	Route tables	Internet Gateway
CIDR awareness	Resource map review	Default network risk analysis

Resume Bullet

RESUME

Explored AWS VPC fundamentals by creating a temporary default VPC, reviewing subnets, route tables, Internet Gateway connectivity, DNS settings, and resource-map relationships, then deleting the VPC and dependent resources after testing.

STAR Interview Story

Situation: The AWS lab environment was ready to begin cloud networking, but first needed hands-on review of how a default VPC is structured.

Task: Create or restore a default VPC for exploration, identify the core networking components, and remove it after testing.

Action: I created a default VPC in us-west-2, reviewed its details, subnets, route table, Internet Gateway, and resource map, then deleted the VPC and five dependent resources.

Result: The lab environment now has documented VPC baseline knowledge without retaining an unnecessary default public network.

Version History

v1.0.0	July 20, 2026	Initial documented implementation. Created and explored a default VPC, reviewed subnets, route table, Internet Gateway, and settings, deleted the default VPC after testing, and embedded provided evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Journey to the Center of the VPC: Getting Started with Cloud Networks.
- User-provided VPC creation and resource map screenshot.
- User-provided VPC deletion confirmation screenshot.
- Project 17 - RegionLockout SCP to restrict activity to approved regions.